Università degli Studi di Padova
Quantum Information with Atoms and Photons

Corso di Masters degree in Physics of Data

Assignment

Cavity blockade: a way to make photons interact

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Contents

1 Assignment **2**

1.1 Explain the physical meaning of H_d. How would you realize such a term? 2

1.2 Rewrite the hamiltonian in the frame of the drive and use the RWA. 2

1.3 Integrate the master equation for enough large times to find the steady state. What is the mean photon number $\langle \hat{a}^\dagger \hat{a} \rangle$? What is the steady-state occupation of the n-photon states? Plot the histogram of occupations and compare with a coherent state distribution. [Suggestion: work in a regime of parameters that allow you to keep the photon Hilbert space sufficiently small (e.g. max $10-50$ photon states) and the average photon number of the steady state small, $\langle \hat{a}^\dagger \hat{a} \rangle \leq 10$. Explain how you made your choice.] 3

1.4 In the presence of a coupling $\tilde{g}$, the previous resonance condition is not valid anymore. Set ω to resonance with the lowest (one-photon) dressed state of the Jaynes-Cummings model and find again the steady state in the strong-coupling condition $\tilde{g} \gg \Gamma, \kappa$ and weak drive $\mathcal{E}$. How important is the value of the pumping strength $\mathcal{E}$? (Be careful that increasing $\mathcal{E}$ too much could lead to instability/bistability) 5

1.5 Can you have occupation of two photon states? If not, why? What happens if you increase the values of the decay rates? And if you change $\tilde{g}$? 5

1.6 A way to understand the suppression of two-photon states is to compute the two-photon correlation function $g^{(2)} \equiv \langle \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} \rangle / \langle \hat{a}^\dagger \hat{a} \rangle^2$, which compares the probability to find two simultaneous photons with the one of having two independent photons at random. In the blockade regime, you should find $g^{(2)} < 1$. 8

1.7 Scan the values of ω around the resonance and graphically represent your findings as in Fig. 2a of the paper by Birnbaum et al. 8

1.8 Can you make analytical predictions for the steady state in a convenient regime? 8

2 Conclusioni **8**

1 Assignment

1.1 Explain the physical meaning of H_d. How would you realize such a term?

The physical meaning of H_d is that of a coherent driving field that pumps the cavity mode at frequency ω. This term represents the interaction between the cavity field and an external classical field, which can be realized experimentally by shining a laser beam into the cavity. The laser frequency ω is typically chosen to be close to the cavity resonance frequency ω_c to efficiently drive the cavity mode. The amplitude $\mathcal{E}$ characterizes the strength of the driving field, which can be controlled by adjusting the intensity of the laser.

1.2 Rewrite the hamiltonian in the frame of the drive and use the RWA.

Consider the resonant case

$$\omega = \omega_a = \omega_c$$

and assume no coupling between the cavity and the atom,

$$\tilde{g} = 0.$$

Given an initially empty cavity,

$$|g, 0\rangle ,$$

we numerically solve the corresponding master equation at resonance (e.g. using the `QuTiP` library).

We assume:

- a small atomic decay rate into modes outside the cavity, denoted by Γ,
- a non-perfect cavity with photon loss rate κ.

A. Initial Hamiltonian ($\hat{H}$) The system Hamiltonian in the laboratory frame consists of the cavity field, a two-level atom, the atom–cavity interaction, and an external driving field:

$$\hat{H} = \hbar\omega_c a^\dagger a + \hbar\omega_a |e\rangle\langle e| + \hbar\tilde{g}\left(\sigma^+ a + \sigma^- a^\dagger\right) + \varepsilon e^{i\omega t} a + \varepsilon^* e^{-i\omega t} a^\dagger. \tag{1}$$

Parameters:

- $\omega = \omega_c = \omega_a$ (resonant driving),
- $\delta \approx 0$ and $\omega \gg \delta$.

B. Unitary Transformation (R) To move to the rotating frame, we define the unitary operator

$$R = e^{i\omega|e\rangle\langle e|t} \otimes e^{i\omega a^\dagger a t} \equiv R_A \otimes R_{\mathrm{EM}}. \tag{2}$$

The transformed Hamiltonian is given by

$$H' = RHR^\dagger - i\hbar R\dot{R}^\dagger. \tag{3}$$

C. Expansion of the Transformation

I. Inertial Term Using

$$R_A \dot{R}_A^\dagger = -i\omega |e\rangle\langle e| , \tag{4}$$

$$R_{\mathrm{EM}} \dot{R}_{\mathrm{EM}}^\dagger = -i\omega a^\dagger a, \tag{5}$$

we obtain

$$-i\hbar R\dot{R}^\dagger = -\hbar\omega a^\dagger a - \hbar\omega |e\rangle\langle e| . \tag{6}$$

II. Transformation of the Hamiltonian Terms The field and atomic operators transform as

$$a_H(t) = R_{\mathrm{EM}} a R_{\mathrm{EM}}^\dagger = a e^{-i\omega t}, \tag{7}$$

$$\sigma_H(t) = R_A \sigma R_A^\dagger = \sigma e^{-i\omega t}. \tag{8}$$

Applying $RHR^\dagger$ term by term:

Cavity Energy

$$1 \otimes R_{\mathrm{EM}}\, \hbar\omega_c a^\dagger a\, R_{\mathrm{EM}}^\dagger = \hbar\omega_c a^\dagger a. \tag{9}$$

Atomic Energy

$$R_A\, \hbar\omega_a \left|e\right\rangle \left\langle e\right|\, R_A^\dagger \otimes 1 = \hbar\omega_a \left|e\right\rangle \left\langle e\right|. \tag{10}$$

Interaction Terms

$$R_A \sigma^+ R_A^\dagger \otimes R_{\mathrm{EM}} a R_{\mathrm{EM}}^\dagger = \sigma^+ e^{i\omega t} \otimes a e^{-i\omega t} = \sigma^+ a, \tag{11}$$

$$R_A \sigma^- R_A^\dagger \otimes R_{\mathrm{EM}} a^\dagger R_{\mathrm{EM}}^\dagger = \sigma^- e^{-i\omega t} \otimes a^\dagger e^{i\omega t} = \sigma^- a^\dagger. \tag{12}$$

Driving Terms

$$\varepsilon e^{i\omega t} a_H(t) = \varepsilon a, \tag{13}$$

$$\varepsilon^* e^{-i\omega t} a_H^\dagger(t) = \varepsilon^* a^\dagger. \tag{14}$$

D. Final Hamiltonian in the Rotating Frame Defining the detunings

$$\delta_c = \omega_c - \omega, \qquad \delta_a = \omega_a - \omega,$$

the transformed Hamiltonian becomes

$$H' = \hbar\delta_c a^\dagger a + \hbar\delta_a \left|e\right\rangle \left\langle e\right| + \hbar g \left(\sigma^+ a + \sigma^- a^\dagger\right) + \xi a + \xi^* a^\dagger. \tag{15}$$

Laboratory Frame Expression For reference, the Hamiltonian in the laboratory frame can be written as

$$H = \hbar\omega_c a^\dagger a + \hbar\omega_a \left|e\right\rangle \left\langle e\right| + \hbar g \left(\sigma^+ a + \sigma^- a^\dagger\right) + \xi a + \xi^* a^\dagger. \tag{16}$$

1.3 Integrate the master equation for enough large times to find the steady state. What is the mean photon number $\langle \hat{a}^\dagger \hat{a} \rangle$? What is the steady-state occupation of the n-photon states? Plot the histogram of occupations and compare with a coherent state distribution. [Suggestion: work in a regime of parameters that allow you to keep the photon Hilbert space sufficiently small (e.g. max $10 - 50$ photon states) and the average photon number of the steady state small, $\langle \hat{a}^\dagger \hat{a} \rangle \leq 10$. Explain how you made your choice.]

Given the conditions of the last point,

- $g = 0$
- $w = w_c = w_a$

we can write the Hamiltonian 15 as the following (using $\hbar = 1$):

$$H' = \xi a + \xi^* a^\dagger \tag{17}$$

As we can see this Hamiltonian is time independent and it represent the continues creation and destruction of photon caused by the pumping field. We now can visualize the evolution of the expected values of the number of photons:

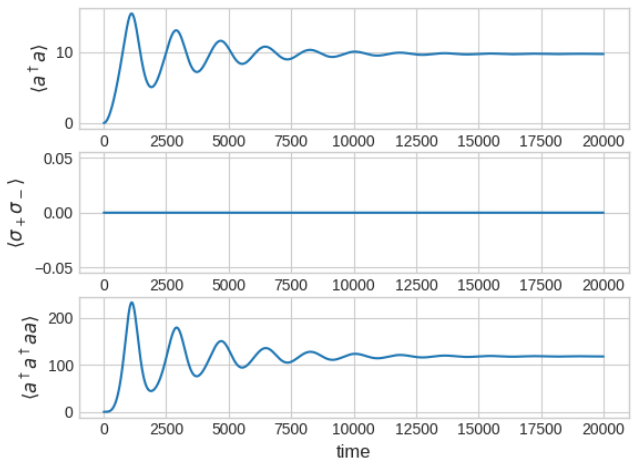

Figure 1: Time evolution of the expected values of the J-C model coupled with a bath

Here we can see that the expected value of finding the atom in the excited state decays exponentially where as the number of photons oscillates reaching an equilibrium by the final time. We also can visualize the different density matrices of the system along with occupation probabilities of each state when the system reach an equilibrium.

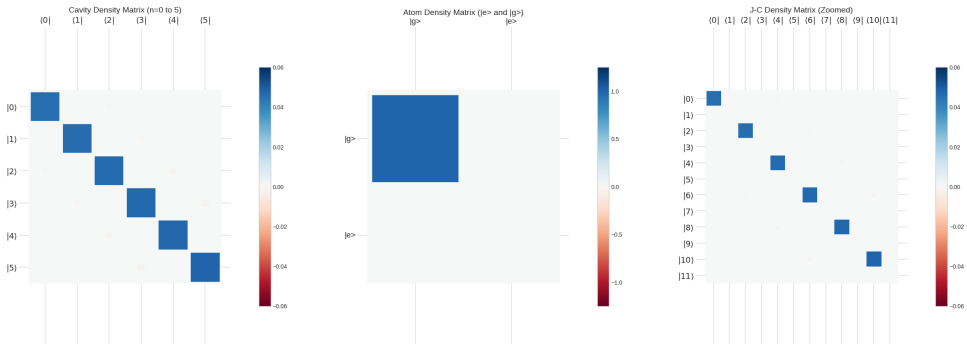

Figure 2: Different density matrices of the system

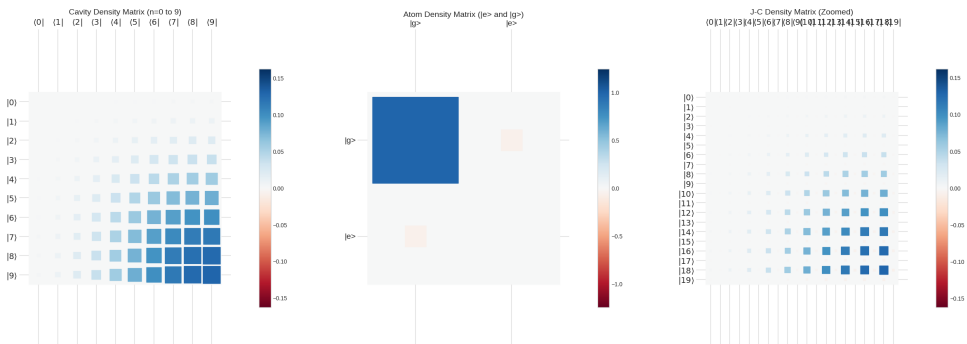

Figure 3: Different density matrices of the system of a coherent state

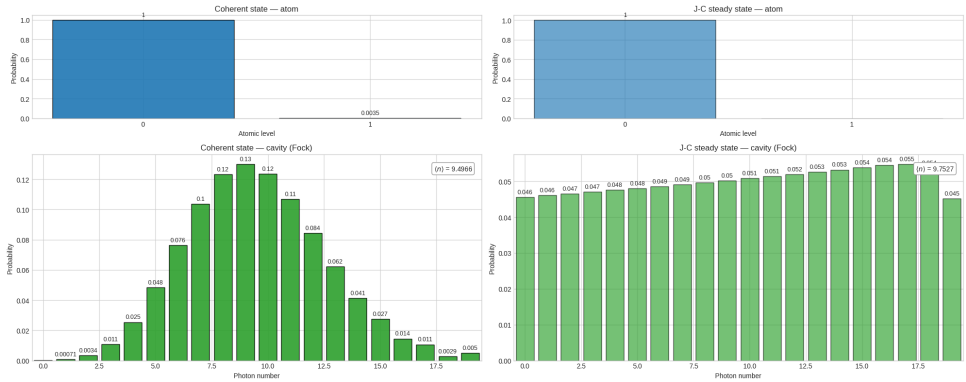

Figure 4: Comparison of the occupation probability of the different states with a coherent state

As we can see the

1.4 In the presence of a coupling $\tilde{g}$, the previous resonance condition is not valid anymore. Set ω to resonance with the lowest (one-photon) dressed state of the Jaynes-Cummings model and find again the steady state in the strong-coupling condition $\tilde{g} \gg \Gamma, \kappa$ and weak drive $\mathcal{E}$. How important is the value of the pumping strength $\mathcal{E}$? (Be careful that increasing $\mathcal{E}$ too much could lead to instability/bistability)

Now we study the coupling effect in the strong coupling regime, in order to compare the results with those provided by the article I have chosen the values $(g, \kappa, \Gamma) = (100, 1, 1)$

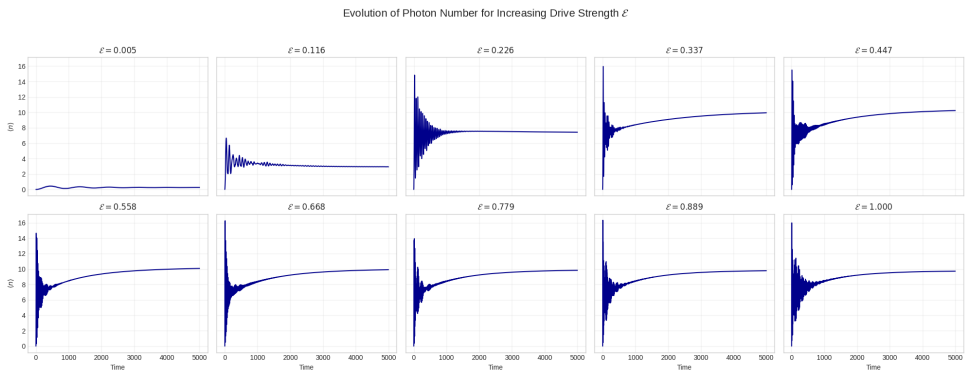

Figure 5: Caption

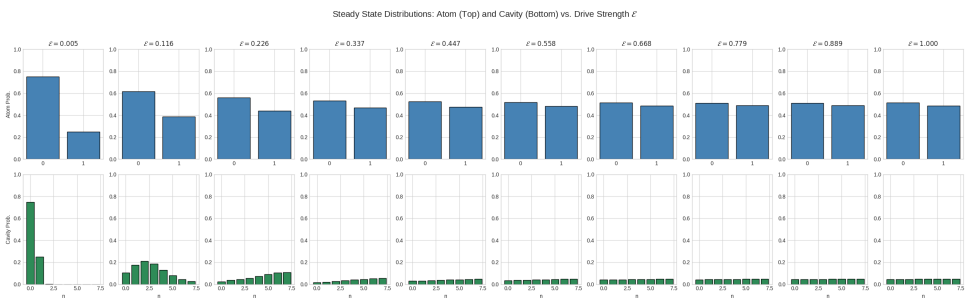

Figure 6: Caption

1.5 Can you have occupation of two photon states? If not, why? What happens if you increase the values of the decay rates? And if you change $\tilde{g}$?

Now we explore the different parameters

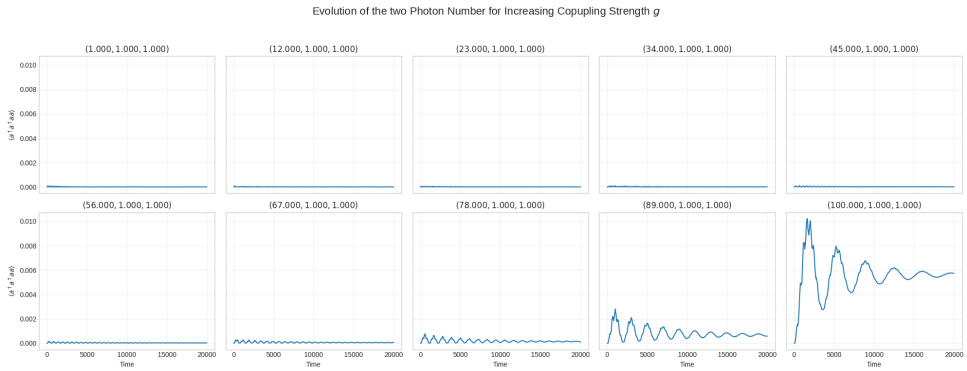

Figure 7: Caption

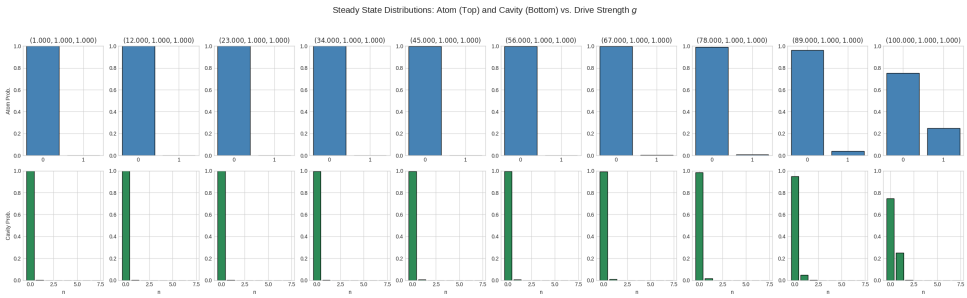

Figure 8: Caption

The coupling parameter $\tilde{g}$

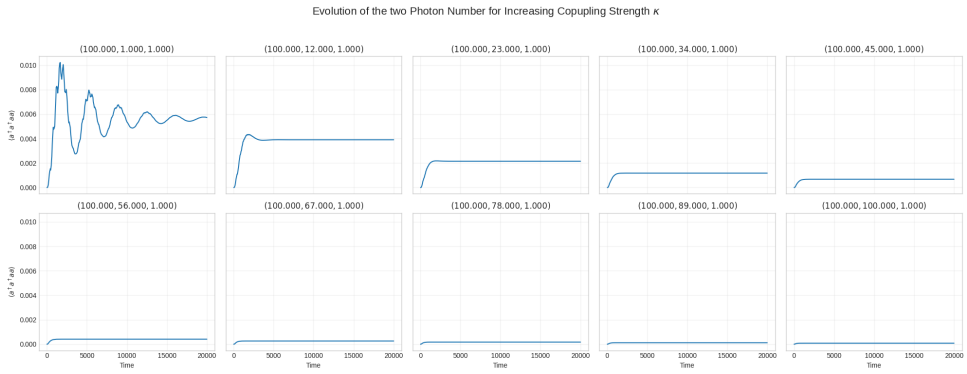

Figure 9: Caption

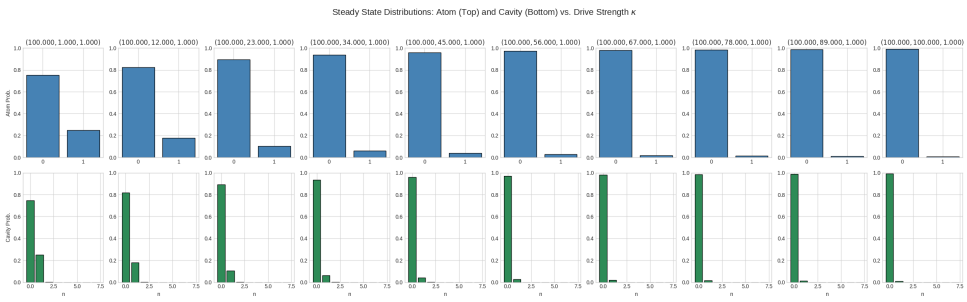

Figure 10: Caption

The coupling parameter κ

Figure 11: Caption

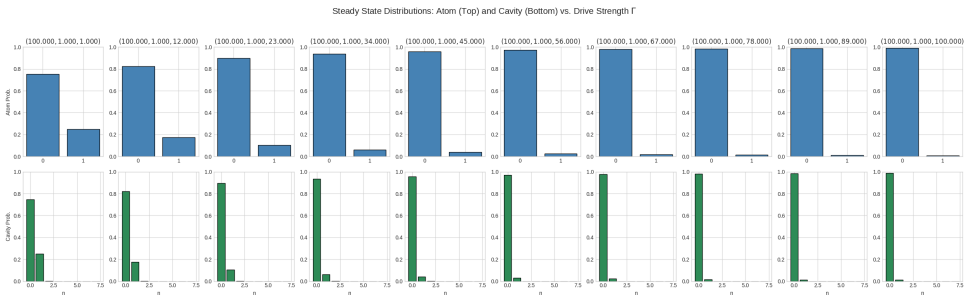

Figure 12: Caption

The coupling parameter Γ

1.6 A way to understand the suppression of two-photon states is to compute the two-photon correlation function $g^{(2)} \equiv \langle \hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} \rangle / \langle \hat{a}^\dagger \hat{a} \rangle^2$, which compares the probability to find two simultaneous photons with the one of having two independent photons at random. In the blockade regime, you should find $g^{(2)} < 1$.

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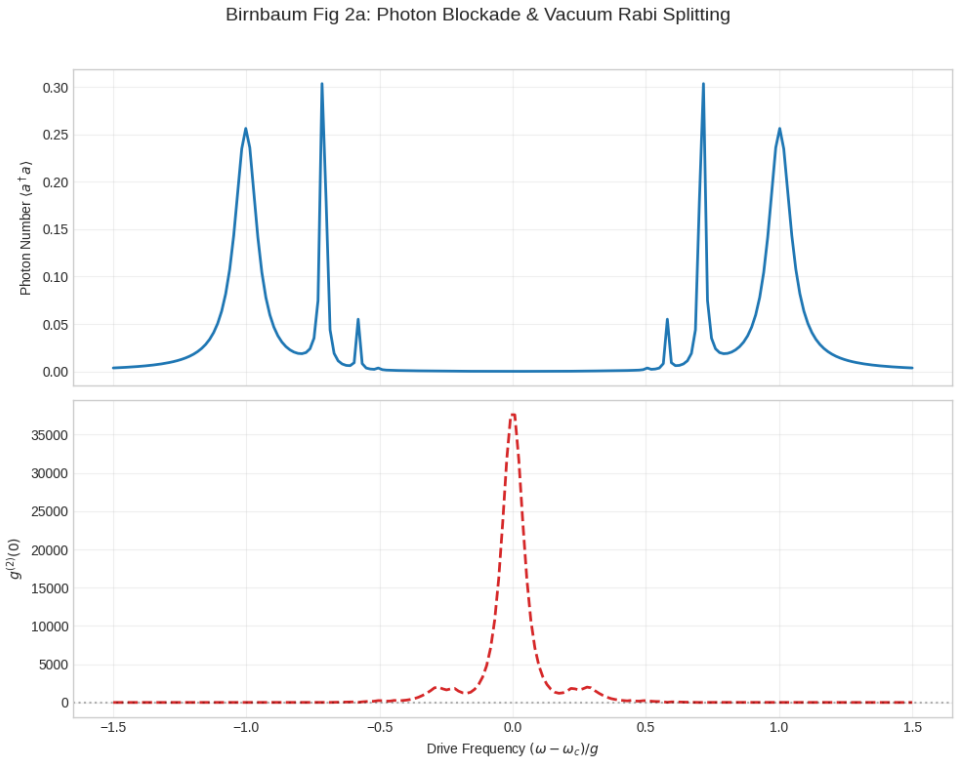

Figure 13: Caption

1.8 Can you make analytical predictions for the steady state in a convenient regime?

In the steady state we have that $\dot{\rho} = 0$ and therefore from the Lindblad master equation we have that:

$$\frac{i}{\hbar}[\hat{H}', \rho] = \kappa \hat{a} \rho \hat{a}^\dagger - \frac{1}{2}\{\hat{a}^\dagger \hat{a}, \rho\} + \Gamma \hat{\sigma}^- \rho \hat{\sigma}^+ - \frac{1}{2}\{\hat{\sigma}^+ \hat{\sigma}^-, \rho\}$$

Where we can define ρ as:

$$\rho = \rho_A \otimes \rho_C \otimes \rho_B = \sigma^- \sigma^+ \otimes$$

2 Conclusioni

Buon lavoro.